

Kinora Hip

Clinical AI Orientation (Part 7)

Structured reference for RAG / AI-assisted physiotherapy consultation

Bones, Labrum, Ligaments, Muscles, Tendons, Bursae, Vascular Supply & Biomechanics

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Disclaimer

*This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established musculoskeletal concepts from standard anatomical texts (Gray's/Standring, Moore, Netter), kinesiology (Neumann), clinical assessment (Magee), and specialty literature (FAI/Warwick Agreement, GTPS/Grimaldi, Doha groin agreement, MCFA/AVN anatomy). Approximate values vary by study. **Red flags — suspected fracture, dislocation, septic arthritis, acute neurovascular compromise, or progressive neurological deficit — require urgent medical/orthopaedic referral.** Femoral head AVN risk counseling applies after fracture/dislocation.*

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1. Hip Bones

RECORD: Femur (Proximal)

ID: HIP-BONE-01

Bone: Proximal femur — femoral head, neck, greater/lesser trochanters, intertrochanteric regions

Landmarks: Femoral head (2/3 sphere; fovea capitis for ligamentum teres); femoral neck (anteversion typically ~10-20 deg adults; neck-shaft angle ~125-135 deg); greater trochanter (glute med/min/piriformis/OI/gemelli insertions; GTPS landmark); lesser trochanter (iliopsoas insertion); intertrochanteric line (anterior — capsule/iliofemoral); intertrochanteric crest (posterior — quadratus femoris); quadrate tubercle; trochanteric fossa; calcar femorale (medial compressive trabeculae)

Muscle Attachments: Gluteus medius/minimus → greater trochanter; piriformis, obturator internus, gemelli → medial/superior GT; obturator externus → trochanteric fossa; quadratus femoris → quadrate tubercle; iliopsoas → lesser trochanter; gluteus maximus → gluteal tuberosity (and ITB); vastus lateralis origin near GT; adductors insert more distal on linea aspera but influence proximal mechanics

Ligament Attachments: Ligamentum teres to fovea capitis; iliofemoral/pubofemoral/ischiofemoral via capsule to intertrochanteric line/crest regions; zona orbicularis encircles neck

Blood Supply: Primary adult femoral head supply: medial circumflex femoral artery (MCFA) deep branch via retinacular (especially posterior-superior) vessels; lateral circumflex femoral (LCFA) lesser contribution; artery of ligamentum teres (obturator) minor in adults, more important in children. Disruption of MCFA retinacular vessels → osteonecrosis risk (fracture, dislocation, surgical approaches)

Biomechanics: Ball-and-socket articulation; neck length/angle creates lever arms for abductors; coxavara decreases abductor MA and may elevate shear; coxavalga opposite; anteversion alters IR/ER arc and patellofemoral tracking cascade

Clinical Importance: Femoral neck fracture (osteoporosis — Garden classification); intertrochanteric fracture; cam morphology (aspherical head-neck — FAI); AVN of femoral head; proximal hamstring avulsion at ischium vs lesser trochanter avulsion (adolescents — iliopsoas); GT apophysitis/avulsion in youth athletes

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Neumann DA. Kinesiology. Gautier E / Kalhor M MCFA anatomy literature (AVN relevance).

RECORD: Acetabulum

ID: HIP-BONE-02

Bone: Acetabulum — hemispheric socket formed by ilium, ischium, and pubis fusion

Landmarks: Lunate (articular) surface; acetabular fossa (non-articular fat/ligamentum teres); acetabular notch inferiorly closed by transverse acetabular ligament; acetabular rim/labrum attachment; anterior wall (pubic contribution); posterior wall (ischial — critical in fracture stability); roof/dome (iliac — primary load); center-edge angle (Wiberg) and acetabular version assessed radiographically

Muscle Attachments: Indirect via capsule and reflected head of rectus femoris (ilioacetabular origin); obturator externus near inferior margin; hip external rotators influence stability; no large muscle belly originates inside the socket

Ligament Attachments: Labrum and transverse acetabular ligament complete the rim; iliofemoral, pubofemoral, ischiofemoral ligaments blend with capsule at rim; ligamentum teres from acetabular notch/transverse ligament to fovea

Blood Supply: Superior gluteal, inferior gluteal, obturator, and MCFA/LCFA anastomotic contributions to acetabular periosteum and walls; relevant in periacetabular osteotomy and fracture fixation

Biomechanics: Covers ~170 deg of femoral head with labrum deepening socket; primary weight-bearing through dome in stance; under-coverage (dysplasia) elevates contact stress and labral load; over-coverage / retroversion associated with pincer FAI

Clinical Importance: Developmental dysplasia of the hip (DDH); acetabular fracture (Letournel/Judet); pincer morphology FAI; osteoarthritis; acetabular labral tears at chondrolabral junction; posterior wall deficiency → instability

References: Standring S. Gray's Anatomy. Ganz R / Clohisy JC FAI literature. Wiberg center-edge angle classic description. Moore KL Clinically Oriented Anatomy.

2. Acetabular Labrum

RECORD: Acetabular Labrum

ID: HIP-LAB-01

Structure: Fibrocartilaginous ring attached to acetabular rim; triangular cross-section; completes socket with transverse acetabular ligament inferiorly; creates suction-seal / fluid seal with femoral head; richly innervated peripherally; relatively hypovascular — healing capacity limited especially in avascular zones

Blood Supply: Radial branches from capsule and periacetabular network (gluteal, obturator, MCFA anastomoses); peripheral labrum better perfused than central free edge — tear location affects healing potential

Innervation: Branches related to nerve to quadratus femoris, obturator, superior gluteal, and direct capsular nerves — explains nociceptive labral pain

Biomechanics: Deepens acetabulum (~21% volume increase cited in anatomic studies); enhances stability via seal and negative pressure; load distribution; proprioception; tear disrupts seal → microinstability and chondral overload

Clinical Tests: FADIR (flexion-adduction-IR) — sensitive but not specific for FAI/labral irritation; FABER — may provoke anterior/lateral pain; log roll; scour/quadrant; resisted SLR; clinical diagnosis requires imaging correlation — no single pathognomonic test

MRI: MR arthrography historically gold-standard for labral tear; high-resolution 3T MRI increasingly sufficient; look for chondrolabral separation, paralabral cyst, cam/pincer morphology, cartilage delamination. Plain radiographs first for FAI bony morphology (Dunn, frog-leg, false-profile, AP pelvis)

Pathologies: Labral tear (anterosuperior most common in cam FAI); degeneration; ossification; paralabral cyst; associated chondral injury; dysplasia-related hypertrophic labrum under shear

Treatment: Activity modification; physical therapy focusing on deep stabilizer control, glute strength, avoiding provocative end-range FAdIR early; NSAIDs as appropriate; image-guided injection selective; arthroscopic repair/debridement ± osteochondroplasty for refractory FAI/labral pathology after failed conservative care

Rehabilitation: Phase protection of repair (surgeon protocol — often limited flexion/IR/rotation early); isometrics → progressive ROM → glute med/max and deep rotator strength → single-leg control → running/cutting return. Non-operative: same kinetic chain without surgical precautions

References: Ganz R et al. FAI concept. Clohisy JC et al. FAI clinical/radiographic. Griffin DR et al. Warwick Agreement on FAI syndrome. Standring S. Gray's Anatomy.

3. Hip Ligaments

RECORD: Iliofemoral Ligament

ID: HIP-LIG-01

Ligament: Iliofemoral ligament (Y ligament of Bigelow) — strongest hip ligament

Origin: AIIS and iliac portion of acetabular rim

Insertion: Intertrochanteric line via two bands (superior/lateral and inferior/medial) — inverted Y

Function: Limits hip extension and external rotation; resists anterior dislocation; allows upright standing with minimal muscle force ('hanging on ligaments')

Biomechanics: Taut in extension; slackens in flexion; primary static restraint to hyperextension; critical for anterior stability with iliopsoas and anterior capsule

Healing: Capsuloligamentous remodeling over months; post-arthroscopy capsular repair protocols protect against extension/ER stress early when capsule closed

Clinical Tests: Passive hip extension end-feel; anterior apprehension patterns; differentiate from psoas pain and anterior labral pain

Treatment: Sprain: protect extension extremes, progressive loading; laxity/microinstability: deepen stabilizer rehab, consider orthopaedic opinion if traumatic

Rehabilitation: Avoid early aggressive hyperextension after anterior capsule injury/surgery; glute max and deep ER strengthening; neuromuscular control in stance

References: Standring S. Gray's Anatomy. Neumann DA. Kinesiology. Myers CA / Torry MR hip capsule biomechanics literature.

RECORD: Pubofemoral Ligament

ID: HIP-LIG-02

Ligament: Pubofemoral ligament

Origin: Iliopubic eminence / superior pubic ramus and obturator crest region

Insertion: Blends with inferior iliofemoral band / capsule toward intertrochanteric region

Function: Limits extension, abduction, and some external rotation; reinforces inferior capsule

Biomechanics: Works with iliofemoral to constrain extension-abduction extremes; inferior capsule important in hyperabduction injuries

Healing: Similar capsular timeline — progressive protected ROM then strength

Clinical Tests: Pain/apprehension in extension-abduction; assess with FABER carefully; inferior instability rare but consider post-trauma

Treatment: Load modification; address adductor-abdominal balance in athletes if coexisting groin pain

Rehabilitation: Graded abduction control; adductor strengthening if weak; avoid early forced hyperabduction after inferior capsule injury

References: Standring S. Gray's Anatomy. Moore KL. Neumann DA. Kinesiology.

RECORD: Ischiofemoral Ligament

ID: HIP-LIG-03

Ligament: Ischiofemoral ligament

Origin: Ischial rim of acetabulum / posterior-inferior capsule origin

Insertion: Blends into zona orbicularis / posterior femoral neck capsule toward GT base

Function: Limits internal rotation and extension; primary posterior capsular restraint to IR

Biomechanics: Taut in extension-IR; contributes to posterior stability; zona orbicularis collar constrains distraction

Healing: Posterior capsule injury after posterior dislocation — protected IR/flexion protocols per trauma team

Clinical Tests: Passive IR in flexion may stress posterior structures; posterior apprehension; differentiate ischiofemoral impingement (QF space) from ligament sprain

Treatment: Post-dislocation: urgent reduction, AVN counseling, protected rehab; isolated sprain uncommon — treat within capsular injury framework

Rehabilitation: Posterior dislocation protocols typically limit flexion/IR/adduction early; progressive abductor and ER strength; gait restoration

References: Standring S. Gray's Anatomy. Neumann DA. Hip dislocation orthopaedic trauma guidelines.

RECORD: Ligamentum Teres

ID: HIP-LIG-04

Ligament: Ligamentum teres (round ligament of femur)

Origin: Acetabular notch and transverse acetabular ligament (bifurcate attachments)

Insertion: Fovea capitis of femoral head

Function: Secondary stabilizer (especially in flexion-ER / extremes in some studies); carries artery of ligamentum teres; proprioceptive role proposed; not primary stabilizer in intact adult bony/labral hip

Biomechanics: Tension patterns debated — may check distraction and certain rotation extremes; more vascular importance in pediatric femoral head

Healing: Tears often managed based on instability symptoms; reconstruction experimental/selected instability cases

Clinical Tests: No reliable isolated bedside test; considered in differential of central groin pain with microinstability after labral/capsule injury

Treatment: Address primary FAI/labral/capsule drivers; arthroscopic debridement vs reconstruction in selected instability — specialist decision

Rehabilitation: Deep hip stabilizer and glute program; avoid early distraction-heavy activities if reconstruction performed (surgeon-guided)

References: O'Donnell J / Martin RL ligamentum teres reviews. Standring S. Gray's Anatomy. Cerezal L imaging of ligamentum teres.

4. Hip Muscles

RECORD: Gluteus Maximus

ID: HIP-MUS-01

Name: Gluteus maximus

Origin: Posterior ilium behind posterior gluteal line, dorsal sacrum/coccyx, sacrotuberous ligament

Insertion: Iliotibial tract (upper fibers ~75%) and gluteal tuberosity of femur (deep/lower fibers)

Innervation: Inferior gluteal nerve (L5-S2)

Blood Supply: Inferior gluteal artery; superior gluteal artery anastomoses

Fiber Direction: Inferolateral from pelvis to ITB/femur — large cross-sectional area

Primary Action: Hip extension; powerful external rotation; posterior pelvic tilt contribution

Secondary Action: Upper fibers assist abduction; lower fibers assist adduction; trunk extension over femur

Stabilizing Function: Primary sagittal and rotary stabilizer in gait propulsion and load transfer; SI force closure via sacrotuberous/thoracolumbar fascia links

EMG: High in sit-to-stand, climbing, sprint acceleration, hip thrust, step-up; lower in quiet standing than deep rotators/abductors for frontal plane

Trigger Points: Buttock referral; may mimic sciatica; sacral and ITB referral patterns

Stretch: Knee-to-chest with adduction variants; figure-4; avoid aggressive neural stretch if sciatic irritation

Strengthening: Hip thrusts, bridges, split squats, step-ups, Romanian deadlift, prone hip extension with knee flexed to bias glute vs hamstring

Clinical Importance: Weakness → hip extensor dominance shift to hamstrings, anterior pelvic tilt patterns, reduced sprint power; relevant in low back and SI rehab

Common Injuries: Strain; contusion (hip pointer adjacent); enthesopathy near origin; post-injection neuropathy rare

Rehabilitation: Pain-free isometrics → bridges → loaded thrusts → unilateral → plyometrics/sprint drills

References: Neumann DA. Distefano / Contreras EMG literature. Standing S. Gray's Anatomy.

RECORD: Gluteus Medius

ID: HIP-MUS-02

Name: Gluteus medius

Origin: External ilium between posterior and anterior gluteal lines

Insertion: Lateral greater trochanter (broad footprint — anterior/middle/posterior facets)

Innervation: Superior gluteal nerve (L4-S1)

Blood Supply: Superior gluteal artery

Fiber Direction: Fan-shaped — anterior, middle, posterior fibers with distinct rotation roles

Primary Action: Hip abduction; pelvic leveling in single-leg stance

Secondary Action: Anterior fibers: flexion/IR; posterior fibers: extension/ER (position-dependent)

Stabilizing Function: Primary frontal-plane pelvic stabilizer; controls femoral adduction/IR

EMG: High in sidelying abduction, single-leg stance, pelvic drop control, single-leg squat; moderate in clamshells

Trigger Points: Lateral hip to buttock and posterior iliac crest referral — GTPS overlap

Stretch: Cross-body adduction in flexion; ITB/glute figure-4 (care if insertional pain)

Strengthening: Sidelying abduction, hip hitches, band walks, single-leg stance endurance, step-downs

Clinical Importance: GTPS/gluteal tendinopathy; Trendelenburg gait; PFP kinetic chain

Common Injuries: Gluteal tendinopathy; partial tears at GT; enthesopathy; superior gluteal nerve injury (iatrogenic)

Rehabilitation: Isometric abduction in slight hip abduction (Grimaldi-informed) → isotonic → functional single-leg → running return; reduce compressive adducted stretching early

References: Grimaldi A gluteal tendinopathy. Distefano LJ JOSPT EMG. Neumann DA.

RECORD: Gluteus Minimus

ID: HIP-MUS-03

Name: Gluteus minimus

Origin: Ilium between anterior and inferior gluteal lines

Insertion: Anterior greater trochanter / capsule

Innervation: Superior gluteal nerve (L4-S1)

Blood Supply: Superior gluteal artery

Fiber Direction: Fan deep to medius — strong anterior IR component

Primary Action: Abduction and internal rotation; femoral head compression into acetabulum

Secondary Action: Assists flexion (anterior fibers)

Stabilizing Function: Intracapsular stabilizer; works with medius in frontal plane

EMG: Co-activates with medius in single-leg and abduction tasks

Trigger Points: Anterior-lateral hip referral into GT and groin occasionally

Stretch: Gentle extension-ER stretch variants — avoid aggressive if tendinopathy

Strengthening: Same closed-chain pelvic control as medius; IR control drills as indicated

Clinical Importance: Often concurrent pathology with medius in GTPS; femoral head stability

Common Injuries: Tendinopathy/tears with medius; enthesopathy

Rehabilitation: Integrated with glute med progressive loading and gait retraining

References: Semciw AI / Grimaldi A gluteal anatomy. Standring S. Gray's Anatomy. Neumann DA.

RECORD: Tensor Fasciae Latae (TFL)

ID: HIP-MUS-04

Name: Tensor fasciae latae

Origin: ASIS and anterior iliac crest

Insertion: Iliotibial band → Gerdy's tubercle (lateral tibia)

Innervation: Superior gluteal nerve (L4-S1)

Blood Supply: Superior gluteal / ascending LCFA anastomoses

Fiber Direction: Vertical anterolateral thigh into ITB

Primary Action: Hip flexion, abduction, IR; tensions ITB

Secondary Action: Assists knee stability via ITB (extension moment when knee near extended)

Stabilizing Function: Anterolateral hip stabilizer; can dominate when glute med weak

EMG: High in flexion-abduction tasks and side-lying clam if form poor (compensation)

Trigger Points: Lateral hip/thigh referral; ITB-related lateral knee pain cascade

Stretch: Standing hip extension-adduction stretch; foam roll ITB adjunctive (evidence mixed)

Strengthening: Usually inhibit dominance — prioritize glute med/max; light abduction only if truly weak

Clinical Importance: Overactivity common in GTPS and PFP; ITB syndrome kinetic contributor

Common Injuries: Strain rare; more often myofascial overuse; snapping hip (external) with ITB

Rehabilitation: Reduce compressive load; strengthen glutes; gait/running cadence; soft-tissue as adjunct

References: Neumann DA. Grimaldi A. Magee DJ Orthopedic Physical Assessment.

RECORD: Iliopsoas

ID: HIP-MUS-05

Name: Iliopsoas (iliacus + psoas major; psoas minor variable)

Origin: Psoas major: T12-L5 vertebral bodies/discs and TPs. Iliacus: iliac fossa. Merge to common tendon.

Insertion: Lesser trochanter of femur

Innervation: Psoas: ventral rami L1-L3 (L2-L4 often cited). Iliacus: femoral nerve (L2-L3)

Blood Supply: Iliolumbar, obturator, external iliac / femoral branches

Fiber Direction: Passes under inguinal ligament over iliopectineal eminence — anterior hip

Primary Action: Primary hip flexor; lumbar stabilizer (psoas); anterior pelvic tilt contribution

Secondary Action: Slight ER of hip; ipsilateral lumbar side-bend (psoas)

Stabilizing Function: Anterior hip stabilizer in extension; lumbar segmental control role debated/nuanced

EMG: High in sit-up/hip flexion; active in late swing of gait to decelerate extension then flex

Trigger Points: Anterior groin, lumbar referral — differentiate from labral/FAI pain

Stretch: Thomas stretch / half-kneeling hip flexor with posterior tilt — avoid lumbar extension substitute

Strengthening: Short-lever hip flexion, standing marches, dead bug progressions with lumbar control; eccentrics for snapping hip when indicated

Clinical Importance: Internal snapping hip (iliopsoas tendon); iliopsoas bursitis; post-THA anterior pain; contribution to lumbar lordosis patterns

Common Injuries: Strain; tendinopathy; lesser trochanter avulsion (youth); bursitis

Rehabilitation: Load management of aggressive hip flexor stretching if insertional pain; progressive eccentric/concentric flexion; address FAI if mechanical snagging

References: Neumann DA. Standring S. Gray's Anatomy. Internal snapping hip clinical reviews.

RECORD: Sartorius

ID: HIP-MUS-06

Name: Sartorius — longest muscle in body

Origin: ASIS

Insertion: Pes anserinus (proximal medial tibia)

Innervation: Femoral nerve (L2-L3)

Blood Supply: Femoral artery branches

Fiber Direction: Oblique across anterior thigh (lateral → medial)

Primary Action: Hip flexion, abduction, ER; knee flexion and tibial IR

Secondary Action: Tailor's position muscle; assists crossing leg

Stabilizing Function: Minor anteromedial dynamic support; more mobility than prime stability

EMG: Active in FABER-like motions and early swing

Trigger Points: Anterior thigh strip referral

Stretch: Extension-adduction-IR combined stretch carefully

Strengthening: FABER against resistance; step-ups; generally not primary rehab target

Clinical Importance: ASIS apophysitis/avulsion in adolescents; pes anserine pain differential

Common Injuries: Proximal avulsion; strain; contusion

Rehabilitation: Progressive loading after apophyseal injury per healing; kinetic chain hip-knee control

References: Standring S. Gray's Anatomy. Moore KL. Magee DJ.

RECORD: Rectus Femoris

ID: HIP-MUS-07

Name: Rectus femoris (only biarticular quadriceps)

Origin: Straight head: AIIS. Reflected head: superior acetabular rim / hip capsule

Insertion: Quadriceps tendon → patella → tibial tuberosity via patellar tendon

Innervation: Femoral nerve (L2-L4)

Blood Supply: LCFA descending branch; femoral perforators

Fiber Direction: Vertical anterior thigh — crosses hip and knee

Primary Action: Hip flexion and knee extension

Secondary Action: Anterior pelvic tilt influence

Stabilizing Function: Anterior hip restraint in extension; knee extensor mechanism

EMG: High in kicking, sprinting, jumping; AIIS stress in kicking sports

Trigger Points: Anterior thigh to knee referral

Stretch: Thomas / prone quad stretch with hip extension — protect lumbar spine

Strengthening: Short-arc quads, Spanish squat variants, Nordic progressions adjunct, kicking drills graded

Clinical Importance: AIIS avulsion; proximal tendinopathy; quadriceps strain (often mid-belly RF)

Common Injuries: Strain grades I-III; AIIS apophyseal injury; contusion/myositis ossificans risk

Rehabilitation: POLICE/PEACE&LOVE principles → isometric → concentric → eccentric → sprint/kick return; imaging if rupture suspected

References: Brukner & Khan. Neumann DA. Standring S. Gray's Anatomy.

RECORD: Adductors (Hip Adductor Group)

ID: HIP-MUS-08

Name: Adductor group — adductor longus, brevis, magnus; gracilis; pectineus (functionally grouped)

Origin: Pubic body/rami and ischiopubic ramus (magnus also ischial tuberosity — hamstring portion)

Insertion: Linea aspera, medial supracondylar line, adductor tubercle (magnus); pes anserinus (gracilis); pectineal line (pectineus)

Innervation: Obturator nerve (L2-L4) primarily; pectineus often femoral; hamstring portion of magnus tibial nerve

Blood Supply: Obturator artery; deep femoral / perforators; MCFA contributions

Fiber Direction: Medial thigh — oblique/vertical to linea aspera

Primary Action: Hip adduction

Secondary Action: Flexion (anterior fibers/pectineus/longus); extension (magnus hamstring part); rotation position-dependent

Stabilizing Function: Medial hip stabilizer in cutting; pelvic ring tension with abdominals (athletic pubalgia continuum)

EMG: High in cutting, side lunges, squeeze drills, skating strides

Trigger Points: Groin to medial thigh referral — Doha groin pain classification relevant

Stretch: Frog / butterfly / lateral lunge stretches — graded in osteitis pubis

Strengthening: Isometric ball squeezes → Copenhagen adduction progression → lateral lunges → sport-specific change of direction

Clinical Importance: Adductor-related groin pain; adductor longus enthesopathy; relationship to osteitis pubis / athletic pubalgia

Common Injuries: Adductor longus strain (most common); proximal enthesitis injury; avulsion rare

Rehabilitation: Doha-guided active rehab; Copenhagen protocol evidence for prevention/rehab; restore side-to-side strength symmetry before RTS

References: Weir A et al. Doha agreement. Thorborg K / Serner A adductor literature. Neumann DA. Standring S.

RECORD: Deep External Rotators

ID: HIP-MUS-09

Name: Deep external rotators — piriformis, gemellus superior/inferior, obturator internus/externus, quadratus femoris

Origin: Pelvic surface of sacrum (piriformis); ischial spine/tuberosity (gemelli/QF); obturator membrane (OI/OE)

Insertion: Medial greater trochanter / trochanteric fossa (group insertions)

Innervation: Nerve to piriformis (S1-S2); nerve to OI/gemellus superior; nerve to QF/gemellus inferior; obturator nerve (OE)

Blood Supply: Inferior gluteal, internal pudendal, MCFA anastomoses

Fiber Direction: Horizontal/oblique from pelvis through sciatic foramina to GT

Primary Action: External rotation (hip near extension); piriformis abducts flexed hip

Secondary Action: Femoral head centering; accessory extension/abduction fiber-dependent

Stabilizing Function: Short rotators compress femoral head — 'rotator cuff of the hip' analogy

EMG: Active in ER and fine rotational control; elevated tone in some pelvic pain cohorts

Trigger Points: Deep buttock; piriformis may refer like sciatica — deep gluteal syndrome differential

Stretch: Figure-4 / FAIR carefully — avoid aggressive stretch if neural irritation or IFI

Strengthening: Band ER, side-lying clam (quality), isometric ER sets, functional rotational control

Clinical Importance: Deep gluteal syndrome; ischiofemoral impingement (QF); pelvic floor sidewall link (OI); SI force closure contributors

Common Injuries: Myalgia/overactivity more than tear; QF edema in IFI; piriformis entrapment variants

Rehabilitation: Differentiate underactive vs overactive; motor control + glute max/med; neural mobility if indicated; image if IFI suspected

References: Martin HD deep gluteal syndrome. Stranding S. Neumann DA. Ischiofemoral impingement MRI literature.

RECORD: Hamstrings (Proximal)

ID: HIP-MUS-10

Name: Proximal hamstrings — biceps femoris long head, semitendinosus, semimembranosus (ischial origin complex)

Origin: Ischial tuberosity (conjoint tendon BF LH + ST; SM separate footprint)

Insertion: Fibular head (BF); pes anserinus (ST); posterior medial tibial condyle (SM) — distal noted for kinetic chain

Innervation: Tibial portion of sciatic nerve (L5-S2) for these proximal hip extensors

Blood Supply: Inferior gluteal; perforating branches of deep femoral

Fiber Direction: Vertical posterior thigh from ischium

Primary Action: Hip extension and knee flexion; decelerate knee extension/hip flexion in swing

Secondary Action: Posterior pelvic tilt; tibial rotation fiber-dependent

Stabilizing Function: Eccentric control in late swing sprinting; sacral force closure via sacrotuberous link

EMG: Very high late-swing sprinting; Nordic curls; RDLs; bridging with knee more extended biases hamstrings

Trigger Points: Ischial and posterior thigh referral — screen sciatic nerve

Stretch: Supine SLR-based hamstring stretch with neural differentiation (slump/SLR)

Strengthening: Isometrics at mid-length → Nordic progression → 45-deg hyperextensions → RDL → sprint drills

Clinical Importance: Most common sprint muscle strain; proximal tendinopathy in runners/hockey; complete proximal avulsion needs urgent ortho opinion

Common Injuries: Myofascial strain (BF LH classic); proximal free-tendon tear; ischial apophyseal avulsion (adolescents)

Rehabilitation: BAGIC-informed imaging when available; progressive eccentric/lengthening; criteria-based RTS (strength symmetry, H-test, sprint mechanics)

References: Askling C hamstring literature. British Athletics BAGIC. Brukner & Khan. Neumann DA.

RECORD: Quadriceps (Hip-Crossing Components)

ID: HIP-MUS-11

Name: Hip-crossing quadriceps components — primarily rectus femoris; vastii influence knee but not hip directly (included for extensor mechanism context)

Origin: AIIS and acetabular rim (RF); vastii from femoral shaft (non-hip-crossing)

Insertion: Quadriceps tendon / patellar tendon complex

Innervation: Femoral nerve (L2-L4)

Blood Supply: LCFA; femoral artery branches

Fiber Direction: Anterior thigh vertical

Primary Action: RF: hip flexion + knee extension; vastii: knee extension only

Secondary Action: Shock absorption in landing via knee; anterior pelvic influence (RF)

Stabilizing Function: Patellofemoral and tibiofemoral control in squat/land; RF anterior hip check

EMG: High in jump landing, cut, squat, kick follow-through

Trigger Points: Anterior thigh to knee

Stretch: Prone quad / couch stretch with posterior pelvic tilt

Strengthening: Squats, step-downs, Spanish squats, terminal knee extension; RF-specific hip flexion when hip flexor weakness isolated

Clinical Importance: Landing mechanics ACL risk context; RF strain in kicking sports; extensor lag after knee injury affecting hip strategy

Common Injuries: RF strain; contusion; tendinopathy at quad tendon (more knee)

Rehabilitation: Progressive closed-chain loading; address hip-knee coupling; sport-specific decelerations

References: Neumann DA. Brukner & Khan. Escamilla squat EMG literature.

5. Tendons

RECORD: Gluteus Medius / Minimus Tendons

ID: HIP-TEN-01

Tendon: Gluteus medius and minimus insertional tendons at greater trochanter

Muscle: Gluteus medius and gluteus minimus

Insertion: Lateral, superoposterior, and anterior facets of greater trochanter (facet-specific footprints)

Blood Supply: Superior gluteal artery periosteal/retinacular branches — insertional zone watershed risk

Biomechanics: Transmit abductor torque; compressive wrapping under ITB in adduction increases tendon load

Common Injuries: Gluteal tendinopathy; partial articular-side tears; full-thickness tears (older adults)

Ultrasound: First-line for insertional thickening, tears, bursal fluid; dynamic assessment possible

MRI: Gold standard for tear grading, muscle atrophy/fatty infiltration, differential GTPS causes

Treatment: Education (avoid aggressive stretching/adduction compression early); isometrics; graded loading; CSI selective; surgery rare for irreparable tears with atrophy

Exercises: Isometric abduction; sidelying abduction with pillow; hip hitches; functional single-leg

Rehabilitation: Grimaldi-style progressive tendon loading 12+ weeks typical; gait retraining

References: Grimaldi A et al. Gluteal tendinopathy. Connell D imaging. Fearon A clinical works.

RECORD: Iliopsoas Tendon

ID: HIP-TEN-02

Tendon: Iliopsoas (conjoint) tendon

Muscle: Iliacus and psoas major

Insertion: Lesser trochanter

Blood Supply: Branches from iliolumbar, obturator, femoral circumflex anastomoses

Biomechanics: Snaps over iliopectineal eminence / femoral head in internal snapping; high flexion loads in sprinting/kicking

Common Injuries: Tendinopathy; internal coxa saltans; post-THA irritation; lesser trochanter avulsion (youth)

Ultrasound: Dynamic visualization of snapping; guided injection into iliopsoas bursa

MRI: Tendon edema, bursitis, lesser trochanter pathology; assess for concurrent FAI labral disease

Treatment: Activity modification; PT; US-guided bursal injection; rare endoscopic release

Exercises: Eccentric hip flexor control; pelvic tilt coordination; graded sprint/kick volume

Rehabilitation: Address lumbar-pelvic control; avoid chronic aggressive end-range stretch if insertional

References: Internal snapping hip reviews. Standring S. Yen YM / Byrd JW hip arthroscopy contexts.

RECORD: Proximal Hamstring Tendons

ID: HIP-TEN-03

Tendon: Proximal hamstring complex (conjoint BF LH–ST and semimembranosus tendons)

Muscle: Biceps femoris long head, semitendinosus, semimembranosus

Insertion: Ischial tuberosity footprints (distinct SM vs conjoint)

Blood Supply: Inferior gluteal and medial circumflex / perforator contributions — watershed near free tendon

Biomechanics: High eccentric load late swing; compressive load in deep hip flexion sitting/stretching

Common Injuries: Proximal tendinopathy; partial tearing; complete avulsion with retraction

Ultrasound: Useful for tendinopathy and some tears; operator-dependent for retraction measurement

MRI: Preferred for tear extent, retraction, sciatic nerve scar relationship

Treatment: Tendinopathy: progressive loading, sit modification. Complete proximal avulsion: early ortho — surgery often if significant retraction/active athlete

Exercises: Isometrics → heavy slow resistance → Nordics/RDL → sprint mechanics

Rehabilitation: Long timeline for proximal free-tendon; criteria-based RTS testing

References: Askling C. Goom T proximal hamstring tendinopathy. BAMIC guidelines.

RECORD: Adductor Longus Tendon / Enthesis

ID: HIP-TEN-04

Tendon: Adductor longus proximal tendon and pubic entheses

Muscle: Adductor longus

Insertion: Pubic body / inferior to pubic crest — fibrocartilaginous entheses

Blood Supply: Obturator and external pudendal / femoral branches

Biomechanics: High load in cutting and skating; shares pubic stress with contralateral adductors/abdominals

Common Injuries: Enthesopathy; acute proximal strain/tear; athletic pubalgia overlap

Ultrasound: Thickening, hyperemia (Doppler), partial fiber disruption

MRI: Pubic bone marrow edema patterns; secondary cleft sign; associated rectus abdominis aponeurosis injury

Treatment: Active rehab per Doha; CSI rarely curative alone; surgery selected chronic cases

Exercises: Copenhagen progression; isometric squeezes; multiplanar cutting graded

Rehabilitation: Restore adductor strength ratio vs abductors; sport-specific COD before RTS

References: Weir A Doha agreement. Thorborg K. Serner A adductor injury MRI studies.

RECORD: Rectus Femoris Proximal Tendons

ID: HIP-TEN-05

Tendon: Rectus femoris direct (AIIS) and reflected (acetabular) heads

Muscle: Rectus femoris

Insertion: Merge into RF muscle then quadriceps tendon distally

Blood Supply: LCFA ascending/descending branches

Biomechanics: Kicking and sprint hip flexion with knee extension — dual tensile demand

Common Injuries: AIIS avulsion; proximal myotendinous strain; reflected head injury near hip

Ultrasound: Dynamic and static assessment of proximal RF; avulsion hematoma

MRI: Defines intramuscular degloving (characteristic RF injury patterns) and apophyseal injury

Treatment: Most strains conservative; displaced AIIS avulsion may need ortho review

Exercises: Graded hip flexion and eccentric quad-hip coupling; kicking volume control

Rehabilitation: Progress through pain-free kicking distances; sprint return last

References: Kassarian A RF injury imaging. Brukner & Khan. Standring S.

6. Bursae

RECORD: Greater Trochanteric Bursa

ID: HIP-BUR-01

Bursa: Trochanteric bursae complex — subgluteus maximus (trochanteric) bursa most discussed; also subgluteus medius/minimus bursae

Location: Between GT / gluteal insertions and ITB—gluteus maximus

Function: Reduce friction of ITB and glute tendons over greater trochanter

Clinical Importance: Pain lateral hip — historically 'trochanteric bursitis'; modern view often gluteal tendinopathy primary with secondary bursal reaction (GTPS)

Common Pathologies: Bursitis secondary to tendinopathy, trauma, or inflammatory disease; calcific rare

Imaging: US: bursal fluid, tendon tears. MRI: bursal T2 signal, tendon pathology, differentiate from referred lumbar pain

Treatment: Load management and tendon loading program primary; short-term NSAID; CSI may help pain but address tendon drivers; avoid sleep-on-side compression

Rehabilitation: Grimaldi GTPS protocol principles; abductor progressive loading; gait/running cues

References: Grimaldi A / Fearon A GTPS. Board TN trochanteric pain reviews.

RECORD: Iliopsoas Bursa

ID: HIP-BUR-02

Bursa: Iliopsoas (iliopectineal) bursa — largest bursa around hip

Location: Between iliopsoas tendon and anterior hip capsule / iliopectineal eminence; may communicate with hip joint in adults (~15% classically cited — variable)

Function: Reduce friction of iliopsoas over anterior capsule and eminence

Clinical Importance: Anterior groin pain differential with labral tear, OA, athletic pubalgia

Common Pathologies: Iliopsoas bursitis; enlargement with RA/OA; internal snapping association; post-THA irritation

Imaging: US-guided assessment/injection; MRI for bursal distension and concurrent joint disease

Treatment: Activity modification; PT; US-guided aspiration/injection; treat underlying FAI/OA

Rehabilitation: Hip flexor load management; pelvic control; graded return to running/kicking

References: Standring S. Gray's Anatomy. Iliopsoas bursitis imaging reviews. Byrd JW hip scopes.

RECORD: Ischial Bursa

ID: HIP-BUR-03

Bursa: Ischial (ischio-gluteal) bursa

Location: Between ischial tuberosity and gluteus maximus — sitting surface

Function: Cushioning during sitting; reduce friction near hamstring origin

Clinical Importance: Weaver's bottom — pain with sitting; differentiate proximal hamstring tendinopathy and pudendal pain

Common Pathologies: Ischial bursitis after prolonged sitting or trauma; chronic sitting occupations/sports

Imaging: MRI/US when needed to differentiate bursitis vs hamstring enthesopathy vs tumor (rare)

Treatment: Sitting load modification (cushion); activity adjustment; rare injection; address training errors

Rehabilitation: Graded sitting tolerance; proximal hamstring loading if concurrent tendinopathy; glute strength

References: Magee DJ. Brukner & Khan. Standring S. Gray's Anatomy.

7. Blood Supply

RECORD: Superior Gluteal Artery

ID: HIP-ART-01

Artery: Superior gluteal artery

Origin: Posterior division of internal iliac artery

Branches: Superficial and deep branches within gluteal region after exiting greater sciatic foramen above piriformis

Structures Supplied: Gluteus maximus (superficial branch); gluteus medius/minimus and TFL (deep branch); hip bone periosteum; contributions to hip joint anastomoses

Venous Drainage: Superior gluteal veins to internal iliac venous system

Clinical Importance: At risk in posterior approaches; landmark with superior gluteal nerve; bleeding source in pelvic trauma; intramuscular injection quadrant awareness

Imaging: CTA/angiography in trauma; Doppler US limited deep gluteal detail

References: Standring S. Gray's Anatomy. Moore KL. Pelvic trauma vascular chapters.

RECORD: Inferior Gluteal Artery

ID: HIP-ART-02

Artery: Inferior gluteal artery

Origin: Anterior division of internal iliac artery

Branches: Muscular branches; anastomotic rami; companion to inferior gluteal nerve/sciatic region

Structures Supplied: Gluteus maximus; short external rotators; upper hamstrings; contributions to cruciate anastomosis of thigh

Venous Drainage: Inferior gluteal veins → internal iliac

Clinical Importance: Exits greater sciatic foramen below piriformis; relevant in deep gluteal surgery and sciatic nerve exposure

Imaging: CTA in pelvic bleeding; anatomic variation awareness

References: Standring S. Gray's Anatomy. Moore KL Clinically Oriented Anatomy.

RECORD: Medial Circumflex Femoral Artery

ID: HIP-ART-03

Artery: Medial circumflex femoral artery (MCFA)

Origin: Typically deep femoral artery (profunda femoris); occasionally femoral artery directly

Branches: Ascending, descending, deep, and transverse branches; deep branch gives retinacular vessels to femoral head (critical posterior-superior retinacular group)

Structures Supplied: Primary blood supply to adult femoral head/neck; adductors; contributions to cruciate anastomosis

Venous Drainage: Medial circumflex femoral veins to deep femoral / femoral veins

Clinical Importance: Injury in femoral neck fracture, hip dislocation, or surgical approaches → avascular necrosis risk; preserve in fracture fixation and arthroplasty planning

Imaging: CTA/MRA when vascularity questioned; MRI for AVN staging (Ficat/ARCO systems)

References: Gautier E et al. Anatomy of MCFA. Kalhor M et al. Femoral head blood supply. Standring S. Gray's Anatomy.

RECORD: Lateral Circumflex Femoral Artery

ID: HIP-ART-04

Artery: Lateral circumflex femoral artery (LCFA)

Origin: Deep femoral artery (usually)

Branches: Ascending, transverse, descending branches

Structures Supplied: Vastus lateralis and anterior thigh; trochanteric region; lesser contribution to femoral head vs MCFA in adults; descending branch used in ALT flap

Venous Drainage: LCFA venae comitantes to profunda/femoral veins

Clinical Importance: Anterior approach and soft-tissue flap planning; trochanteric anastomosis participant; descending branch landmark in anterolateral thigh

Imaging: Doppler/CTA for flap perforators; trauma angiography

References: Standring S. Gray's Anatomy. Moore KL. Plastic surgery ALT flap anatomy texts.

RECORD: Obturator Artery

ID: HIP-ART-05

Artery: Obturator artery

Origin: Anterior division of internal iliac (typical); may anastomose with inferior epigastric (corona mortis)

Branches: Anterior and posterior branches around obturator foramen; acetabular branch → artery of ligamentum teres

Structures Supplied: Adductor compartment; obturator externus; acetabulum; femoral head contribution via ligamentum teres (minor in adults, important in children)

Venous Drainage: Obturator veins to internal iliac

Clinical Importance: Corona mortis variant — hemorrhage risk in pelvic surgery/hernia repair; acetabular fracture fixation bleeding

Imaging: CTA to map corona mortis before pelvic ring/acetabular surgery

References: Standring S. Gray's Anatomy. Corona mortis anatomic studies. Moore KL.

RECORD: Femoral Artery

ID: HIP-ART-06

Artery: Femoral artery (common femoral at hip region)

Origin: Continuation of external iliac artery distal to inguinal ligament

Branches: Superficial epigastric, superficial circumflex iliac, superficial/deep external pudendal; profunda femoris (deep femoral) arises in femoral triangle giving MCFA/LCFA/perforators

Structures Supplied: Anterior thigh via profunda; superficial branches to skin/inguinal region; entire lower limb downstream

Venous Drainage: Femoral vein (medial to artery in mid triangle — NAVEL mnemonic at inguinal)

Clinical Importance: Pulse landmark midway ASIS–pubic symphysis; catheterization access; injury in penetrating trauma; relationship to iliopsoas and femoral nerve in triangle

Imaging: Duplex US; CTA runoff; angiography

References: Standring S. Gray's Anatomy. Moore KL. Vascular surgery access anatomy.

8. Hip Biomechanics

RECORD: Hip Flexion

ID: HIP-BIO-01

Movement: Hip flexion

Plane: Sagittal

Axis: Mediolateral through femoral head

Primary Movers: Iliopsoas, rectus femoris, sartorius, TFL

Secondary Movers: Adductor longus/pectineus (from extension), anterior glute med/min

Stabilizers: Deep rotators for head centering; abdominals for pelvic control

Arthrokinematics: Femoral head spins posteriorly relative to acetabulum (convex-on-concave); inferior glide accessory with flexion

Osteokinematics: 0 to ~120 deg pure flexion (soft-tissue dependent); more with knee flexed (RF slack)

Joint Reaction Forces: JRF rises with active flexion against load; anterior capsule slackens

EMG: High iliopsoas/RF EMG in marches, kicks, high knees

Torque: Hip flexor MA greatest mid-range; RF dual hip-knee torque tradeoff

Compensations: Lumbar extension substitution; anterior pelvic tilt; hip hike

Common Dysfunctions: FAI pinching in deep flexion+IR; iliopsoas tendinopathy; flexion contracture

Functional Activities: Sit-to-stand initiation, stair ascent, swing phase

Sport Applications: Sprinting knee drive, kicking cocking, hurdling

Assessment: Thomas test; AROM/PROM flexion; FADIR for symptom reproduction

Corrective Exercises: Dead bugs, marches with posterior tilt, graded hip flexor loading, mobility if true length deficit

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Hip Extension

ID: HIP-BIO-02

Movement: Hip extension

Plane: Sagittal

Axis: Mediolateral through femoral head

Primary Movers: Gluteus maximus, hamstrings (esp. magnus hamstring part)

Secondary Movers: Posterior glute med; adductor magnus

Stabilizers: Abductors for frontal control; core for pelvic neutral

Arthrokinematics: Femoral head spins anteriorly; superior accessory tendencies with end-range

Osteokinematics: ~10-20 deg beyond neutral typical; limited by iliofemoral ligament

Joint Reaction Forces: High JRF in terminal stance and loaded hinge; posterior structures taut

EMG: High glute max EMG in thrusts, sprint propulsion, climbing

Torque: Glute max large CSA — high extension torque capacity

Compensations: Lumbar extension; hamstring dominance; knee flexion substitute

Common Dysfunctions: Extensor weakness; hanging on Y-ligament; sprint-related hamstring risk

Functional Activities: Gait propulsion, rising from chair, climbing

Sport Applications: Sprinting, jumping takeoff, Olympic lifts lockout

Assessment: Prone extension MMT; bridge quality; gait terminal stance

Corrective Exercises: Hip thrusts, RDLs, step-ups, sprint mechanics drills

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Hip Abduction

ID: HIP-BIO-03

Movement: Hip abduction

Plane: Frontal

Axis: Anteroposterior through femoral head

Primary Movers: Gluteus medius, gluteus minimus, TFL

Secondary Movers: Piriformis (flexed hip), sartorius, upper glute max

Stabilizers: Contralateral trunk musculature in closed chain; QL may hike pelvis

Arthrokinematics: Femoral head rolls superior / glides inferior (convex-on-concave)

Osteokinematics: ~40-45 deg typical open-chain abduction

Joint Reaction Forces: Single-leg stance JRF often 2.5-4x BW classically cited (Pauwels/Bergmann ranges vary)

EMG: High glute med EMG in sidelying abduction and pelvic drop control

Torque: Abductor torque critical to counter contralateral pelvic drop moment

Compensations: Hip hike (QL); TFL dominance; trunk lean

Common Dysfunctions: Trendelenburg; GTPS; contralateral pelvic drop in running

Functional Activities: Single-leg stance, side-stepping, getting out of car

Sport Applications: Cutting, skating, lateral agility

Assessment: Trendelenburg sign; sidelying MMT; single-leg stance endurance

Corrective Exercises: Hip hitches, sidelying ABD, band walks, step-downs

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Hip Adduction

ID: HIP-BIO-04

Movement: Hip adduction

Plane: Frontal

Axis: Anteroposterior through femoral head

Primary Movers: Adductor magnus, longus, brevis; gracilis; pectineus

Secondary Movers: Lower glute max; QF; hamstrings (accessory)

Stabilizers: Ipsilateral abductors eccentrically control; pelvic ring abdominal coupling

Arthrokinematics: Femoral head rolls inferior / glides superior

Osteokinematics: ~20-30 deg past midline typical

Joint Reaction Forces: High adductor demand in COD; pubic stress with repetitive load

EMG: High adductor EMG in Copenhagen, squeeze, skating

Torque: Adductor longus common strain site under eccentric lengthening

Compensations: Scissoring gait; crossing midline collapse in squat

Common Dysfunctions: Adductor-related groin pain; osteitis pubis continuum

Functional Activities: Crossing legs, side-lying bed mobility, horse riding

Sport Applications: Soccer kick plant, skating push recovery, rugby scrum binding

Assessment: Squeeze tests; Copenhagen capacity; Doha groin classification

Corrective Exercises: Isometric squeezes → Copenhagen progression → COD drills

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Hip Internal Rotation

ID: HIP-BIO-05

Movement: Hip internal rotation

Plane: Transverse

Axis: Longitudinal through femoral head/shaft

Primary Movers: Anterior glute med/min, TFL; adductors position-dependent

Secondary Movers: Semitendinosus/semimembranosus at knee influence limb IR cascade

Stabilizers: Deep ERs eccentrically control IR in closed chain

Arthrokinematics: Spin with small accessory glides; capsule posterior structures restrain IR in flexion

Osteokinematics: ~30-40 deg typical (large individual variation; anteversion dependent)

Joint Reaction Forces: IR in flexion increases anterior/superior labral stress in cam FAI

EMG: TFL/anterior glute EMG in IR tasks

Torque: IR torque smaller than ER group aggregate in many positions

Compensations: Femoral IR collapse (dynamic valgus) in landing

Common Dysfunctions: FAI pain with FAdIR; excessive IR from anteversion

Functional Activities: Pivoting, change of direction

Sport Applications: Golf follow-through, dance turnout opposite demand, cutting

Assessment: Seated/prone IR ROM; FAdIR; single-leg step-down IR quality

Corrective Exercises: Deep ER strength; glute med posterior fibers; landing mechanics

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Hip External Rotation

ID: HIP-BIO-06

Movement: Hip external rotation

Plane: Transverse

Axis: Longitudinal through femoral head/shaft

Primary Movers: Gluteus maximus, piriformis, OI, gemelli, QF, obturator externus

Secondary Movers: Posterior glute med; sartorius

Stabilizers: Abductors and pelvic floor sidewall (OI) for combined stability

Arthrokinematics: Spin; anterior capsule/iliofemoral restrain ER extremes

Osteokinematics: ~40-60 deg typical (position and bony version dependent)

Joint Reaction Forces: ER strength protects against dynamic valgus; deep ERs center head

EMG: High deep ER and glute max EMG in clam/ER band work and cutting

Torque: Glute max large ER torque contributor especially in extension

Compensations: Lumbar rotation substitute; hip flexion substitute in clam

Common Dysfunctions: Deep gluteal pain; ER weakness with ACL/PFP risk cascade

Functional Activities: Getting out of car, pivoting

Sport Applications: Dance turnout, martial arts kicks, cutting sports

Assessment: Prone ER MMT; sidelying clam quality; FABER ROM

Corrective Exercises: Band ER, clam progressions, glute max hinges, rotational control chops

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Squat

ID: HIP-BIO-07

Movement: Squat pattern (bilateral sit-to-stand loading)

Plane: Multiplanar (dominant sagittal)

Axis: Instantaneous hip-knee-ankle axes

Primary Movers: Glute max, quadriceps; hamstrings co-contract

Secondary Movers: Adductors in deep squat; erectors for trunk

Stabilizers: Glute med for knee alignment; core canister

Arthrokinematics: Increasing hip flexion with femoral posterior spin/relative glide needs; depth limited by FAI/bony morphology

Osteokinematics: Hip flexion, knee flexion, ankle DF coupled

Joint Reaction Forces: Hip JRF substantial at depth; varies with trunk lean and bar position

EMG: Quad vs glute bias depends on depth and trunk angle

Torque: Extensor torques shared hip-knee; moment arms shift with depth

Compensations: Knee valgus; heel rise; lumbar flexion; hip shift asymmetry

Common Dysfunctions: FAI pinching deep squat; adductor strain; posterior chain weakness

Functional Activities: Chair rise, toileting, lifting preparation

Sport Applications: Weightlifting, team sport strength training

Assessment: Bodyweight squat screen; overhead squat; pain mapping depth

Corrective Exercises: Box squat to tolerable depth; tempo; glute bias hinges; mobility only if true limit

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Deadlift

ID: HIP-BIO-08

Movement: Deadlift / hip hinge

Plane: Sagittal dominant

Axis: Hip mediolateral axis primary

Primary Movers: Glute max, hamstrings, adductor magnus; erector spinae isometric

Secondary Movers: Lats for bar path; quads off floor

Stabilizers: Abdominal brace; scapular stability

Arthrokinematics: Hip flexion-extension arc with relatively stable knees vs squat

Osteokinematics: Hip hinge osteokinematics — vertical shin more than squat

Joint Reaction Forces: High hip extensor demand; lumbar shear if rounded under load

EMG: High posterior chain EMG; technique-dependent lumbar erector load

Torque: Hip extension torque primary after bar passes knees

Compensations: Lumbar flexion; bar drift; hyperextension lockout

Common Dysfunctions: Hamstring strain risk with poor progression; lumbar pain from hinge fault

Functional Activities: Picking objects from floor

Sport Applications: Powerlifting, strongman, field sport strength

Assessment: Hip hinge screen; RDL capacity; pain with load

Corrective Exercises: Dowel hinge, RDL, block pulls, tempo deadlifts

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Running

ID: HIP-BIO-09

Movement: Running

Plane: Multiplanar

Axis: Cyclical multi-joint axes

Primary Movers: Glute max propulsion; iliopsoas swing; glute med stance stability

Secondary Movers: Hamstrings late-swing deceleration; adductors COD

Stabilizers: Pelvic floor/core canister; foot-ankle stiffness cascade

Arthrokinematics: Rapid flexion-extension with rotation and slight abd/add oscillations

Osteokinematics: Flight phase + stance; step rate/length tradeoff

Joint Reaction Forces: Peak hip contact forces often several $\times$ BW (instrumented prosthesis data vary by speed)

EMG: Burst patterns per phase — see literature phase plots

Torque: High power requirements with speed; abductor moments each stance

Compensations: Overstride; contralateral pelvic drop; hip IR collapse; crossover gait

Common Dysfunctions: GTPS, proximal hamstring tendinopathy, FAI flare, stress fracture risk

Functional Activities: Fitness running, field locomotion

Sport Applications: All running sports; sprint events highest hamstring demand

Assessment: Gait video; pelvic drop; cadence; pain localization

Corrective Exercises: Cadence cues, run-walk, hip abductor endurance, graded exposure

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Walking

ID: HIP-BIO-10

Movement: Walking gait — hip contribution

Plane: Multiplanar

Axis: Cyclical

Primary Movers: Iliopsoas swing flexion; glute max early stance; abductors midstance

Secondary Movers: Hamstrings terminal swing; adductors brief

Stabilizers: Glute med critical single-limb support

Arthrokinematics: Extension to ~ 10 deg terminal stance; flexion ~ 30 deg swing

Osteokinematics: Normal gait ROM substantially less than available PROM

Joint Reaction Forces: Hip JRF $\sim 2.5\text{--}4\times$ BW typical walking citations (varies)

EMG: Phasic low-moderate EMG vs running

Torque: Abductor moment sustains contralateral pelvis

Compensations: Trendelenburg; vaulting; shortened step length; antalgic lean

Common Dysfunctions: OA limited extension; abductor weakness; leg-length compensation

Functional Activities: Community ambulation, ADLs

Sport Applications: Base for return-to-run progressions

Assessment: Observational gait; timed walking; Trendelenburg

Corrective Exercises: Gait drill, assistive device as needed, abductor endurance, extension mobility

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Single-Leg Stance

ID: HIP-BIO-11

Movement: Single-leg stance

Plane: Frontal dominant + multiplanar sway

Axis: Hip joint center

Primary Movers: Gluteus medius/minimus isometric/eccentric
Secondary Movers: TFL, upper glute max, deep rotators
Stabilizers: Trunk position modulates demand — contralateral load increases abductor demand
Arthrokinematics: Minimal osteokinematic motion if stable; micro-adjustments continuous
Osteokinematics: Pelvis level target; femur stacked over foot
Joint Reaction Forces: High sustained JRF — classic teaching model for cane use (ipsilateral cane reduces demand)
EMG: High continuous glute med EMG
Torque: Abductor torque $\approx$ body weight $\times$ lever to contralateral COM line
Compensations: Trunk lean over stance; pelvic drop; toe gripping
Common Dysfunctions: GTPS, Trendelenburg gait, PGP, balance disorders
Functional Activities: Dressing, stairs pause, reaching
Sport Applications: Skating, kicking plant limb, landing hold
Assessment: SLS time/quality; pelvic landmarks; eyes open/closed
Corrective Exercises: Hip hitches, SLS holds, step-downs, suitcase carries
References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Jumping

ID: HIP-BIO-12
Movement: Jump takeoff
Plane: Sagittal dominant
Axis: Multi-joint
Primary Movers: Glute max, quads, plantarflexors triple extension
Secondary Movers: Hamstrings co-contraction; adductors
Stabilizers: Core stiffness for force transfer
Arthrokinematics: Rapid hip extension from flexed countermovement
Osteokinematics: Countermovement then propulsion
Joint Reaction Forces: Peak forces high — multiple $\times$ BW
EMG: Explosive glute/quad EMG burst
Torque: Hip extension power key to jump height
Compensations: Excessive forward lean; knee dominant only; asymmetric push
Common Dysfunctions: Proximal hamstring or RF strain if unprepared
Functional Activities: Jumping to reach, hop sports
Sport Applications: Basketball, volleyball, athletics
Assessment: CMJ / hop testing when appropriate
Corrective Exercises: Progressive plyometric ladder after strength base
References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Landing

ID: HIP-BIO-13
Movement: Jump landing / deceleration
Plane: Multiplanar
Axis: Multi-joint
Primary Movers: Eccentric quads, glutes; hip extensors absorb
Secondary Movers: Hamstrings; abductors control valgus

Stabilizers: Trunk and foot position dictate hip demand

Arthrokinematics: Rapid flexion with controlled IR/abd moments

Osteokinematics: Soft landing increases hip-knee flexion excursion

Joint Reaction Forces: Very high transient JRF and soft-tissue loads

EMG: High eccentric EMG in glute med/max and quads

Torque: Frontal-plane moments linked to ACL injury risk cascade

Compensations: Stiff landing; dynamic valgus; contralateral pelvic drop

Common Dysfunctions: ACL risk pattern; GTPS flare; patellofemoral pain

Functional Activities: Drop landing, cutting deceleration

Sport Applications: All jump-land sports; change of direction

Assessment: Drop-jump assessment; 2D frontal plane projection angle

Corrective Exercises: Soft-land cues, hip strategy, single-leg land progressions

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

RECORD: Kicking

ID: HIP-BIO-14

Movement: Kicking (instep / side-foot models)

Plane: Multiplanar — sagittal dominant swing limb

Axis: Hip and knee axes

Primary Movers: Iliopsoas and RF swing limb acceleration; plant-limb glute med stability

Secondary Movers: Adductors in side-foot; trunk rotators

Stabilizers: Plant-limb single-leg stance stabilizers critical

Arthrokinematics: Swing hip rapid flexion with rotation; end-range stress on AHS/RF

Osteokinematics: Approach angle changes plane demands

Joint Reaction Forces: High swing-limb velocity — RF and hip flexor peak loads

EMG: RF and iliopsoas high toward ball contact; plant limb abductor high

Torque: Large hip flexion torque on swing limb; large frontal torque on plant limb

Compensations: Lumbar side-bend; plant-foot crash; poor follow-through control

Common Dysfunctions: RF strain, AHS apophysitis, adductor strain, plant-limb GTPS

Functional Activities: Ball striking in soccer/football/martial arts

Sport Applications: Soccer, rugby place kicking, martial arts

Assessment: Kicking pain map; plant-limb SLS; RF strength/flexibility

Corrective Exercises: Graded kicking distances; RF and adductor capacity; plant-limb strength

References: Neumann DA. Kinesiology. Magee DJ. Brukner & Khan. Escamilla / Distefano EMG as applicable.

9. Evidence and Guidelines

RECORD: Key Evidence and Guidelines — Hip Module

ID: HIP-EVID-01

Topic: Foundational references for Kinora hip AI orientation

Anatomy: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Atlas.

Biomechanics: Neumann DA. Kinesiology of the Musculoskeletal System. Instrumented hip force literature (Bergmann et al.).

FAI / Labrum: Ganz R et al. FAI. Griffin DR et al. Warwick Agreement on FAI syndrome. Clohisy JC radiographic standards.

GTPS: Grimaldi A et al. Gluteal tendinopathy / GTPS clinical and loading evidence.

Groin: Weir A et al. Doha agreement meeting on terminology and definitions of groin pain in athletes.

Vascular / AVN: Gautier E, Kalhor M et al. MCFA and femoral head blood supply anatomic studies.

Assessment: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

AI Use Note: ROM values, JRF multiples, and test properties are approximate and study-dependent. Prefer current imaging/surgical protocols and individual assessment over static recall.

References: See fields above.